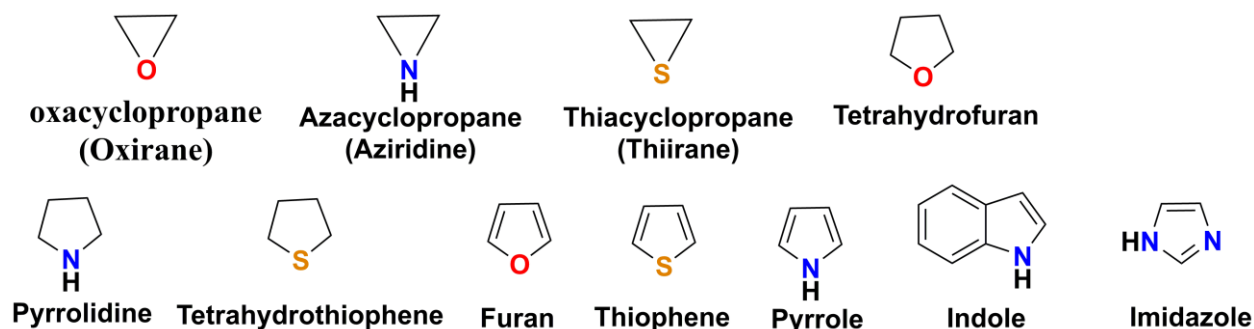


Experiment 6

Preparation of Benzimidazole

Heterocyclic compounds (Heterocycles) are cyclic compounds with one or more atoms of the ring are atom other than carbon. The name comes from the Greek word *heteros*, which means “different”. A variety of atoms such as N, O, S, Se, P and As can be incorporated into the ring structure. Heterocycles are important class of compounds and include many important drugs, and most vitamins.

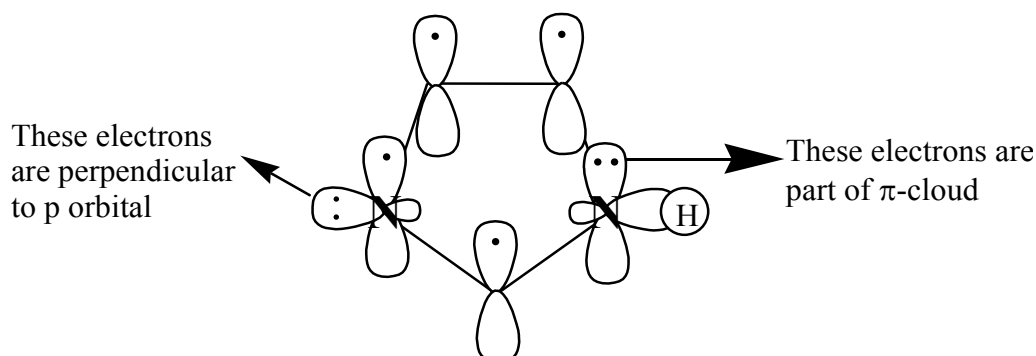
Saturated heterocyclic compounds do not contain double bond in its structure. A saturated heterocycle can be named as cycloalkane after prefixing “aza” for nitrogen, “Oxa” for oxygen and “Thia” for sulfur.



Pyrrole, furan and thiophene are five-membered unsaturated heterocyclic compounds and each has three pairs of delocalized π -electrons and is aromatic.

Indole, benzofuran and benzothiophene contain a five-membered aromatic ring fused to a benzene ring. Indole, benzofuran and benzothiophene are aromatic because they are cyclic planar, every atom has a p-orbital and has five pairs of π -electrons.

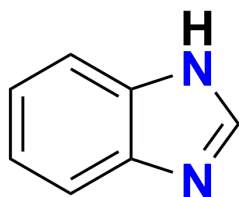
Imidazole is an aromatic compound with three pairs of π -electrons. The resonance energy of imidazole is 14 kcal/mol, less than that of benzene (36 kcal/mol). Imidazole can be protonated because the lone pair of one nitrogen is in the sp^2 orbital. It can exist in both protonated and unprotonated form at physiological pH and that is one of the reasons why histidine, an imidazole containing amino acid is an important catalytic component of many enzymes.



Orbital structure of imidazole

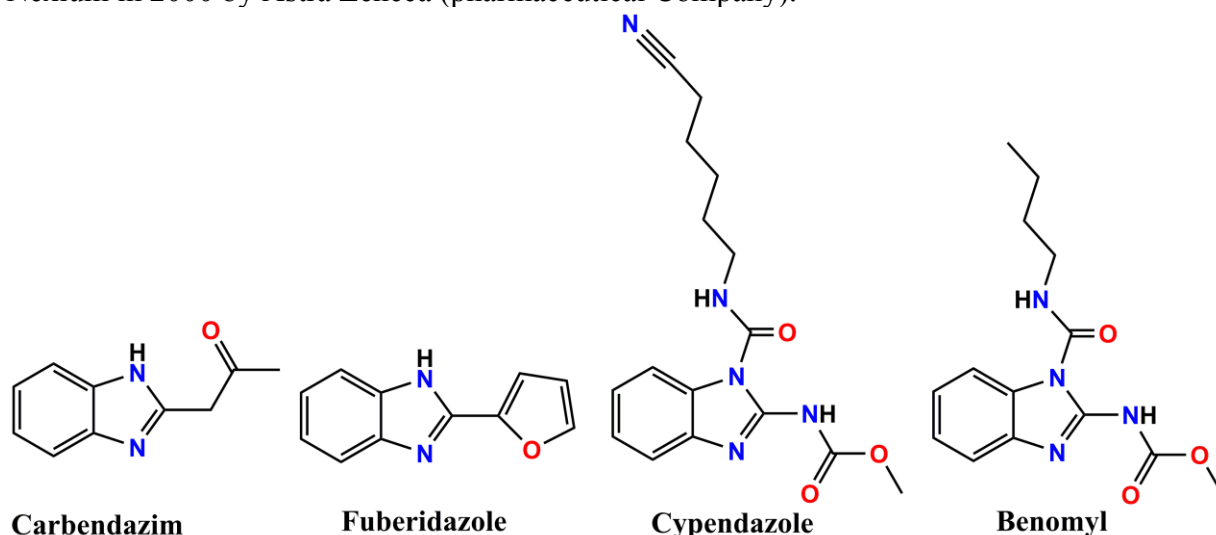
Benzimidazole contains an imidazole ring fused to a benzene ring. Benzimidazole and its derivatives are used as vermicides or fungicides as they inhibit the action of certain microorganisms. Examples of benzimidazole class fungicides include benomyl,

carbendazim, chlorfenazole, cypendazole, debacarb, fuberidazole, furophanate, mecarbinzid, rabenzazole, thiabendazole, thiophanate.

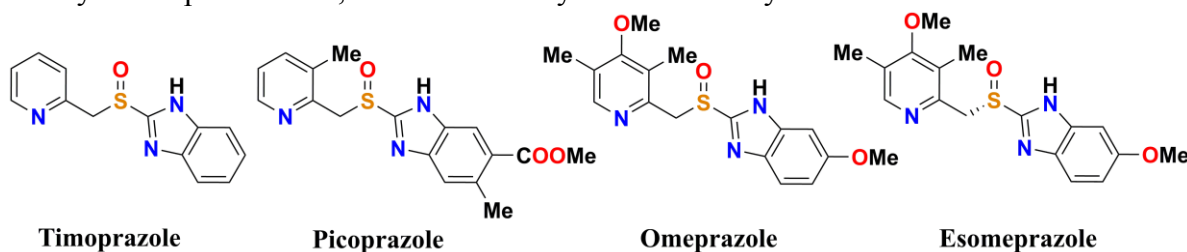


Chemical structure of benzimidazole

Benzimidazole structure is the nucleus in some drugs such as proton pump inhibitors and anthelmintic agents. Three decades ago, disorders associated with inappropriate levels of gastric acid were a major problem and for which treatment options were limited. The medical treatment for gastric acid related disorders in particular peptic ulcers and gastroesophageal reflux disease (GERD) had a breakthrough with the discovery of “Omeprazole”, a benzimidazole based drug. Omeprazole is the first proton pump inhibitor used in clinical practice. Omeprazole is a racemate consisting of two optical isomers (enantiomers). The S isomer-esomeprazole-is superior to Omeprazole both as a gastric acid inhibitor and for the clinical management of GERD. Esomeprazole was launched as Nexium in 2000 by Astra Zeneca (pharmaceutical Company).

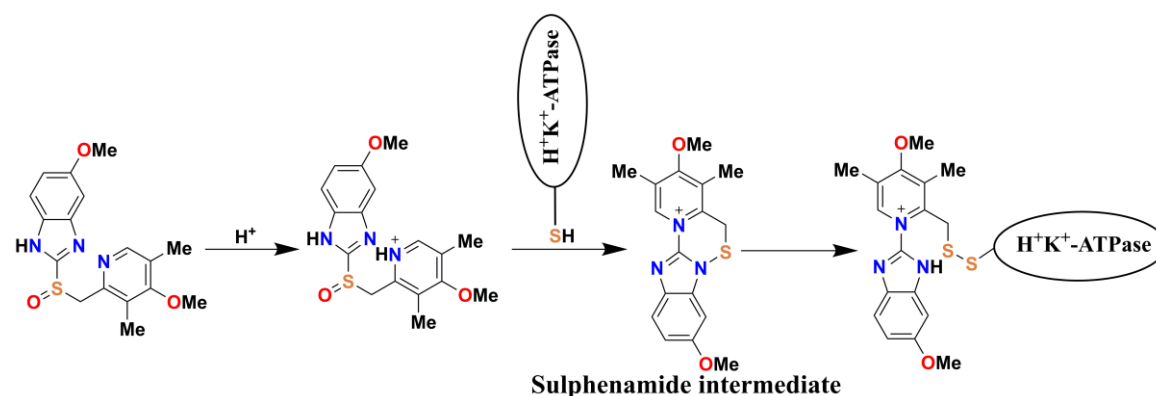


Omeprazole and related proton pump inhibitors share a benzimidazole frame work. These are pro-drugs, since these are stable at neutral pH and rearrange in acidic medium to form tetracyclic sulphenamides, which covalently inhibit the enzyme.



Chemical milestones in the development of proton-pump inhibitors

It was shown in a thorough mechanistic investigation that the higher pKa value of the Omeprazole pyridine ring increased the rate of acid-mediated conversion to active suphenamide species, which is the major factor determining the inhibitory activity.



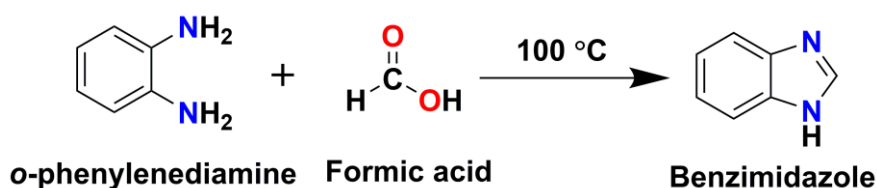
Omeprazole is a weak base, and specifically concentrates in the acidic secretory parietal cells, where it is activated by a proton-catalyzed process to generate sulphenamide. The sulphenamide interacts covalently with the sulfhydryl groups in the extra cellular domain of the enzyme.

In this experiment, we will prepare benzimidazole. This unit is part of variety of biologically relevant molecules.

Chemicals required:

o-Phenylenediamine: 2 g
Formic acid (90%): 2 mL
10% NaOH solution

PROCEDURE



In a 150 mL conical flask fitted with an air condenser, take 2 g of o-phenylenediamine and 2 mL formic acid. Place the flask in a water bath and heat for 1 hr at 100°C (use this time to take the melting point of o-phenylenediamine). Cool the reaction mixture and adjust the pH to alkaline (monitor the pH of the solution by using pH paper) by drop wise addition of 10% aq. NaOH with constant shaking. Filter the precipitate thus obtained by using a Büchner funnel with suction. Wash the precipitate with ice cold water three times. Dry the crude product (first by pressing with the filter paper and then keep it in the oven for sometime) and take its weight. Dissolve one third of the crude product in 30 mL boiling water. Allow it to cool to room temperature and then cool in ice bath (*please do not keep hot beaker in the ice bath*), when pure benzimidazole separate out (note 1 and 2). *Filter it through a fluted filter paper (note 3), dry and take the melting point.*

Notes

1. The crude benzimidazole is yellow tinged. This discoloration is difficult to remove and persists after two crystallizations. This discoloration can be removed by treatment with animal charcoal (*not performed here*).
2. The recrystallization begins immediately on cooling. The filtration must be rapid.
3. *Fluted filter paper is used to separate a liquid and a solid. This arrangement of folds in the filter paper will allow the liquid to pass through it very quickly and give more surface area on which to collect the crystals.*

RESULTS

Total amount of benzimidazole obtained	=	g
Percentage Yield	=	%
Melting point of o-phenylenediamine	=	°C
Melting point of Benzimidazole	=	°C

For further Reading

Organic syntheses, Coll. Vol. 2, p. 12